

BRIAN JARRETT

Senior Backend & Platform Engineer

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PROFESSIONAL SUMMARY

Highly analytical engineering professional with a 30-year foundation in complex system validation, data integrity, and autonomous technology adaptation. Proven track record of entering unfamiliar technical ecosystems, rapidly mastering complex architectures, and becoming a Subject Matter Expert (SME) for the organization's technology stack. Expert in diagnosing legacy infrastructure, verifying data quality at the source to prevent the propagation of bad data and ensure the integrity of the final output, and architecting robust ETL pipelines that merge disparate inputs into high-integrity sources of truth. An objective, narrative-free troubleshooter who translates intricate system data into actionable executive metrics, applying pragmatic innovation—including targeted AI/LLM workflows—strictly where it maximizes operational value.

SKILLS

Data & Systems Engineering:

- Python (Pandas, NumPy, Flask, SQLAlchemy)
- ETL Pipeline & Data Validation
- PostgreSQL, MySQL, MSSQL
- Validation and QA Testing
- Systems/Robotics Design

Systems & Distributed Scale:

- Cisco UCS & Fabric Interconnect
- OpenStack (Nova, Heat)
- vSphere/ESXi, XenServer, Proxmox/ KVM
- Ceph Block Storage (RBD)
- High-Availability (HA) Clustering

DevOps, Security & Leadership:

- Ansible, SaltStack
- CI/CD (GitHub Actions, Jenkins)
- FreeIPA/LDAP, RBAC/HBAC
- FedRAMP Compliance, Qualys
- Monitoring, Diagnostics, Troubleshooting
- FIRST Robotics Mentoring & Leadership

PROFESSIONAL EXPERIENCE

Cisco Systems | Cloud Engineering Technical Leader

Aug 2019 - July 2026 (Remote) * DevSecOps & Compliance Automation: Engineered Python tools parsing raw **STIG XCCDF (XML)** data with the **Qualys API** to generate audit-ready checklists, eliminating hundreds of manual audit hours. * **Data Platform Engineering:** Designed, deployed, and operated a vulnerability management system using a custom ETL pipeline to extract data from **Jira API** and synchronize it to a **PostgreSQL** database. Built a **Python (Pandas, SQLAlchemy)** web frontend and connected **Tableau** dashboards to display real-time SLA metrics. * **Zero-Day Incident Response (Log4j):** Authored enterprise **Ansible** playbooks during the 2021 Log4j event to perform deep filesystem audits across server fleets, locating dormant nested instances that bypassed standard **Qualys** scans and auditing 100% remediation. * **Technical Guidance & Mentorship:** Provided technical guidance across federal cloud environments, mentoring multi-tier engineering teams in methodical troubleshooting and complex vulnerability remediation under strict SLAs. * **Skills:** Python (Pandas, SQLAlchemy), PostgreSQL, REST APIs (Jira, Qualys), ETL Pipelines, Tableau, Ansible, RHEL/Ubuntu, Cisco UCS, STIG/FedRAMP Moderate Compliance.

Cisco Systems | Cloud Systems Engineer

Jan 2017 - Aug 2019 (Remote) * Enterprise Identity Architecture: Designed and deployed a **High-Availability (HA) FreeIPA cluster** using **Ansible** to implement RBAC/HBAC across **2,000+ servers, storage nodes, and network devices**, replacing an aging 389-ds directory framework. * **Virtualized Infrastructure Operations:** Administered large-scale compute environments using **vSphere** on **Cisco UCS** hardware. Orchestrated **Qualys** vulnerability scanning and patch remediation across commercial and FedRAMP sites while maintaining core bastion and identity services. * **Skills:** Ansible, FreeIPA, OpenStack, VMware vSphere, Cisco UCS, RHEL/CentOS, Windows Server, RBAC/HBAC, Qualys, FedRAMP Compliance.

Select Group @ Cisco | Cloud DevOps/QA Engineer (Contract)

Feb 2016 - Jan 2017 (Remote) * Performance Optimization: Identified and removed redundant **Nova Scheduler** filters in **OpenStack**, reducing instance scheduling time for 700 nodes from 20 seconds to 5 seconds

(a **75% improvement** in orchestration speed). * **Infrastructure-as-Code at Scale:** Configured and evaluated **RHUI (Red Hat Update Infrastructure)** and **Proxmox** clusters for InterCloud Services. Automated bare-metal server configuration using **Ansible** via lights-out management (LOM) interfaces. Troubleshooted and resolved a critical OpenStack **cloudbase-init** race condition on Windows images that was causing constant reboots. * **Cloud QA & Orchestration:** Managed global test cycles for Compute and Heat components during cloud upgrades. Developed **Python**, **Bash**, and **Ansible** scripts to perform comprehensive automated orchestration tests. * **Skills:** OpenStack, Ansible, Python, Bash, RHUI, Proxmox, CentOS/RHEL/Windows, Performance Tuning, Cloud QA.

Australian Laboratory Services | Senior Systems & Storage Administrator

Sep 2014 - Jan 2016 (Grand Junction, CO) * **Distributed Storage & Virtualization:** Engineered an Ubuntu-based **Ceph cluster** for **VMware ESXi** hosts, benchmarking iSCSI vs. RBD performance. Used **Ceph RBD snapshots** for automated schema migration testing with zero errors. * **Infrastructure Automation:** Wrote custom **Python (SQLAlchemy)** verification scripts to audit SQL foreign keys during data migration. Implemented **SaltStack** to automate Linux VM and Ceph node provisioning. * **System Stability & Monitoring:** Integrated **Nagios** and **Observium** to monitor network operations, reducing service restarts. Diagnosed WAN/VPN bottlenecks and upgraded gateway hardware to double network throughput. * **Skills:** Ubuntu, Ceph RBD, VMware ESXi, SaltStack, Python, SQLAlchemy, Nagios, Observium, MySQL, Asterisk, VPN Routing.

Garfield County School District 16 | Senior Systems/Network Administrator

Oct 2002 - Jun 2014 (Parachute, CO) * **Systems Integration & Automation:** Programmed a custom **Flask (SQLAlchemy)** app syncing Student Info Systems with **Active Directory** and **Google Apps**, reducing account helpdesk tickets by **97%**. * **Enterprise Modernization & Development:** Replaced legacy databases with centralized PostgreSQL platforms, and automated state compliance reporting via custom PHP scripts. * **Infrastructure & Systems Virtualization:** Centralized desktop computing with **Windows Terminal Servers** and **HP Thin Clients** for 6 labs, saving **\$60k+** in CAPEX. * **Network Engineering:** Designed a high-speed WAN using **Ruckus wireless bridges**, increasing bandwidth by **6,500%** (1.5Mbps to 100+ Mbps) with zero monthly service costs. Deployed district-wide Cisco CallManager. * **Skills:** Python, Flask, SQLAlchemy, PHP, PostgreSQL, XenServer, VirtualBox, Cisco Routers/Switches/PIX, Cisco CallManager, Ruckus Wireless.

Foundational Experience | Systems Engineering, Software QA & Avionics Foundations

1991 - 2002 (Various) * **Solutions Engineer/Software QA Lead (StorageSoft/Phoenix Tech):** Gathered requirements, deployed software, and provided training for customers. Managed testing and support departments. Structured QA testing workflows for **ImageCast** disk cloning tools and designed Windows/Linux/Networking test lab environments. * **Airborne Radar Journeyman (U.S. Air Force):** Repaired high-power microwave systems, high-voltage circuits, and analog/digital systems on AWACS aircraft. * **Skills:** Analog & Digital Diagnostics, PC Architecture, Disk Cloning (ImageCast), Software QA, Network Design, Technical Training.

EDUCATION & CERTIFICATIONS

- **Linux Academy / A Cloud Guru:** AWS & Red Hat Enterprise Linux Curriculum
- **Regis University:** C Programming & Unix Systems Courses
- **Lincoln Christian University:** Music Theory & Composition
- **Colorado School of Mines:** 2 Years - Electrical Engineering & Applied Physics
- **Community College of the Air Force:** Avionics/Electronics Technician - RADAR Specialist (AWACS)
- **Field Training Detachment - Tinker AFB, OK:** AWACS RADAR and IFF Systems Journeyman - Awarded Senior Airman Below-the-Zone
- **Air Force Technical Training School - Keesler AFB, MS:** Electronics and AWACS RADAR Courses - Honor Graduate (99% Academic Test Average)

For a list of legacy certifications, see [linkedin.com/in/bjarrett/details/certifications/](https://www.linkedin.com/in/bjarrett/details/certifications/)